

Thomas Bainbridge

Portfolio\Website

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PROFESSIONAL SUMMARY

A technically proficient postgraduate engineering student with strong interests in machine learning and mathematical modelling, whilst remaining informed on wider areas of engineering. Knowledge has been developed through extensive and in-depth study at both undergraduate and postgraduate level, supported by group and independent research projects. Strongly motivated and committed to delivering robust engineering outcomes while maintaining high professional standards.

EDUCATION

Cranfield University

MSc Computational Fluid Dynamics (2025-2026) Distinction (Predicted)

Key Modules: Numerical Modelling for Incompressible and Compressible Flows, Turbulence Modelling, Data Analysis and Uncertainty, Grid Generation, Numerical Methods and High-Performance Computing, Fluid Mechanics and Heat Transfer

University of Hertfordshire

BEng Automotive Engineering with Motorsport (2021-2025) Upper Second Class Honours (Achieved)

Key Modules: Fluid Mechanics, Thermodynamics, Lean Manufacturing and Supply Chains, High Performance Engine Design, Automotive Electronics and Control Systems, Materials in Engineering, Renewable Energy and the Environment

KEY SKILLS

Software: ANSYS Fluent, Star CCM+, OpenFOAM, Pointwise, ParaView, Tecplot 360, Catia V5, SOLIDWORKS, Fusion 360, OpenVSP, FreeCAD, Pi Toolbox, Linux Operating System, Microsoft Office

Programming: Fortran, MATLAB, Simulink, Python, PyFluent, PyTorch, C++, Julia

TECHNICAL PROJECTS AND EXPERIENCE

Neural networks and Universal Differential equations for UAV parameter estimation from flight test data, Industrial Thesis in partnership with SAAB Bluebear (In Progress):

- Developing a Julia/SciML-based framework for UAV parameter estimation from real flight-test data using Universal Differential Equations and Neural ODE methods.
- Building physics-informed dynamic models that combine flight-dynamics equations with data-driven components to capture nonlinear aerodynamic and propulsion coupling.
- Processing and validating flight-test datasets in collaboration with SAAB BlueBear to assess the predictive capability and engineering usefulness of surrogate UAV models.

Design and Optimisation of Multi-Purpose Defence Drone Group Project, Sponsored by VisionWave (2026):

- Contributed to the design and optimisation of a multi-purpose UAV to satisfy two distinct and conflicting mission requirements, supporting concept development and overall vehicle performance assessment.
- Developed automated CFD and optimisation workflows to evaluate UAV design iterations efficiently, while also serving as group leader to coordinate meetings and track weekly progress.
- Drone concept has been selected to be prototyped as decided by the sponsor.

UH Racing Formula Student Team (2022-2025):

- Year 1: Produced CAD models and detailed technical drawings to support manufacture and assembly of race-car components.
- Year 2: Joined the Aerodynamics department, contributing to the design and evaluation of key aero components while maintaining documentation for the team lead.
- Year 3: Appointed Vice Aerodynamics Lead; managed project planning to meet pre-competition deadlines (Silverstone) and deputised for the Aero Lead in cross-team design reviews, proposing design updates and implementation changes.

PREVIOUS EMPLOYMENT

Aramark, Hospitality Assistant (Part Time) 3 Years

- Trusted with key handling, opening/closing duties, and maintaining high customer service standards.